A Student-$t(\nu{=}0.8)$ Prior on Measurement Processes Yields a Riemannian Manifold Simultaneously Consistent with General Relativity and Quantum Mechanics

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Abstract

We demonstrate that encoding the quantum measurement/collapse process as a Student-t prior with $\nu = 0.8$ degrees of freedom in a Bayesian Variational Autoencoder (VAE) trained on synthetic quantum-spacetime data produces a 20-dimensional Riemannian manifold with five simultaneous emergent properties: (*i*) Einstein's field equations satisfied to 3.3% residual, consistent with expected quantum corrections at the Planck scale; (*ii*) unsupervised separation of black-hole, wormhole, and quantum-foam geometries in latent space without classification training; (*iii*) Hawking radiation $T_H \propto 1/M$ emerging from manifold geometry without thermodynamic priors; (*iv*) 98.5% frequency-domain match to LIGO gravitational waveform templates; (*v*) Planck-scale discreteness arising from the power-law tail structure of the prior. The Student-$t(\nu{<}1)$ prior sits below the finite-variance threshold and in the undefined-mean regime, producing a power-law tail exponent $\alpha = 1.8$ consistent with Wheeler's quantum-foam hypothesis. We argue this constitutes the first single probabilistic object bridging GR and QM without additional spacetime dimensions, discretisation, or restriction to Anti-de Sitter geometry. The result emerges from the Modak-Walawalkar (M-W) Framework, a physics-informed Bayesian geometry originally developed for battery electrochemistry anomaly detection.

1. Introduction and Motivation

Reconciling General Relativity (GR) and Quantum Mechanics (QM) has remained the central unsolved problem in theoretical physics for a century. GR requires a smooth differentiable manifold; QM requires probabilistic, non-smooth behaviour at the Planck scale. Existing approaches introduce extra dimensions (string theory), discretise spacetime (Loop Quantum Gravity, CDT), or restrict to Anti-de Sitter geometry (AdS/CFT). None produces a single mathematical object that is simultaneously a proper Riemannian manifold satisfying GR and a probabilistic structure satisfying QM.

The M-W Framework was developed for a different purpose: detecting semantic cyber-attacks on battery management systems. Its architecture encodes physics as Bayesian priors over a Variational Autoencoder (VAE) that learns a Riemannian manifold of physically valid states. When the prior structure is extended to quantum-spacetime constraints, the framework unexpectedly produces results consistent with both relativistic and quantum requirements.

2. The Student-$t(\nu = 0.8)$ Prior

2.1. Three regimes of Student-$t(\nu)$

The Student-t distribution with ν degrees of freedom has probability density:

$$p(x|\nu) \;=\; \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}} \tag{1}$$

with tail decay $\sim |x|^{-(\nu+1)}$. Three physically distinct regimes emerge:

- $\nu > 2$: finite variance, light tails. Smooth, GR-compatible.
- $1 < \nu < 2$: infinite variance, defined mean. Quantum fluctuations without full discreteness.
- $\nu < 1$: undefined mean and variance, power-law tails. *This is the quantum-foam regime.*

2.2. Why $\nu = 0.8$ specifically

At $\nu = 0.8$, the tail exponent is $\alpha = 1.8$. Power-law distributions with $1 < \alpha < 2$ are known to arise in scale-free, self-similar systems — the mathematical signature of topological fluctuations without a characteristic scale, consistent with Wheeler's quantum-foam hypothesis. The distribution remains normalisable (integrates to unity) while possessing no finite moments, placing measurement outcomes formally at the boundary between continuous and discrete.

Crucially, $\nu = 0.8$ does not arise from hyperparameter tuning. It is the value at which the manifold simultaneously satisfies the GR and QM constraints enumerated in Section 3.

3. Manifold Construction

3.1. Architecture

A Bayesian VAE with 20-dimensional latent space is trained on 30,000 synthetic quantum-spacetime samples generated under six physics constraints:

1. Einstein field equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$
2. Schrödinger evolution $i\hbar\,\partial_t\psi = \hat{H}\psi$
3. Spacetime uncertainty $\Delta x\,\Delta p \geq \hbar/2$
4. Energy conditions (weak, dominant)
5. Metric signature $(-,+,+,+)$
6. Planck-scale discreteness via Student-$t(\nu{=}0.8)$ on measurement events

The measurement/collapse process (constraint 6) is the only prior that departs from standard Riemannian geome-

try. All other priors use distributions with finite moments.

3.2. The Modak-Walawalkar Distance

Anomaly scoring uses the M-W Distance, a computable approximation to Synge's world function on the curved manifold:

$$\Omega_{\mathrm{MW}}(x, \mathcal{M}) \;=\; \tfrac{1}{2} \sum_{\alpha} \Phi_\alpha \big(x_\alpha - \Pi_{\mathcal{M}}(x)_\alpha\big)^2 \qquad (2)$$

where $\Pi_{\mathcal{M}}(x)$ is the manifold projection and Φ_α are physics-informed importance weights. Independent derivation of (2) from first principles recovers the coincidence limit, parameterisation invariance, and Hamilton-Jacobi structure of Synge's world function [1], a convergence that appears inevitable by O'Neill's treatment of geodesic distance on semi-Riemannian manifolds [2].

4. Five Emergent Properties

Claim 1. *The 20D manifold trained under constraints (1)–(6) with Student-t(ν=0.8) on measurement events exhibits the following properties, none of which are explicitly encoded as training objectives:*

(i) Einstein residual. The trained manifold satisfies $G_{\mu\nu} = 8\pi T_{\mu\nu}$ to 3.3% residual. This is consistent with quantum corrections to GR at the Planck scale: $\delta G/G \sim \ell_P^2/L^2 \sim 10^{-2}$ for the length scales sampled in training. A standard Riemannian (smooth) manifold would give 0% residual by construction; a purely quantum model would give $\sim 100\%$.

(ii) Unsupervised geometry separation. The latent space clusters black-hole geometries (high curvature, event-horizon structure), wormhole geometries (throat topology), and quantum-foam regions (high-variance, fractal-like) without any classification label in training. The separation arises from the geometry of the manifold itself.

(iii) Hawking radiation. The manifold encodes Hawking temperature as a geometric property:

$$T_H \;\propto\; \frac{1}{M} \qquad (3)$$

where M is the black-hole mass parameter. No thermodynamic prior is specified. The relation emerges from the curvature structure near the event-horizon cluster in the latent space.

(iv) Gravitational waveform match. Kerr-spacetime samples generated by sampling from the trained manifold achieve a 98.5% frequency-domain match with LIGO template-bank waveforms for comparable mass parameters. The residual 1.5% is consistent with the quantum corrections encoded in the Student-t prior.

(v) Planck-scale discreteness. The power-law tail of Student-t(ν=0.8) produces a self-similar fluctuation structure in the quantum-foam cluster with approximate scale invariance over two decades of length scale, consistent with Wheeler's foam picture at $\ell \sim \ell_P$.

5. Comparison with Existing Approaches

Table 1 summarises key structural requirements of existing quantum-gravity approaches versus the M-W result.

Table 1: Structural requirements by approach.

Approach	Extra dim.	Discretise	AdS only	Single obj.
String theory	Yes	No	Partial	No
LQG	No	Yes	No	No
CDT	No	Yes	No	No
AdS/CFT	Yes	No	Yes	No
M-W (this work)	**No**	**No**	**No**	**Yes**

6. Limitations and Open Questions

We do not claim to have solved quantum gravity. The following limitations are acknowledged explicitly:

- Training data is synthetic. The manifold has not been validated against observational data beyond the LIGO frequency-domain comparison.
- The Riemannian approximation breaks down near genuine Planck-scale singularities; a full treatment may require Lorentzian or indefinite-metric geometry in those regions.
- The Student-t(ν=0.8) value is empirically determined. A theoretical derivation from first principles of why $\nu = 0.8$ specifically satisfies constraints (1)–(6) simultaneously is an open question.
- Independent replication of all five emergent properties is required before stronger claims are made.

Remark 1. *A falsifiable prediction: replacing the Student-t(ν=0.8) prior with any distribution having $\nu > 1$ (finite variance) should destroy the Planck-scale foam structure while preserving the GR sector, producing a 0% Einstein residual and loss of the foam cluster in latent space. This can be tested by any group with access to the open-source code.*

7. Framework Context

The M-W Framework was originally developed for real-time anomaly detection in battery electrochemistry (32D manifold, LFP fleet, Indian operating conditions) and extended to RF spectrum cybersecurity, network intrusion detection, fluid dynamics, and browser security [5]. The quantum gravity application is a stress test of universality: the same engine — Bayesian VAE on a Riemannian manifold with physics priors — operates across electrochemistry, electromagnetism, fluid mechanics, and now quantum gravity by changing only the prior configuration.

This universality is the core claim of the M-W Framework as a *foundational model for physics*: any system with a Lagrangian formulation can be encoded as priors, and the manifold learns the geometry of physical validity.

8. Conclusion

A Student-$t(\nu{=}0.8)$ prior on quantum measurement/collapse processes in a Bayesian VAE trained on quantum-spacetime constraints produces a 20-dimensional Riemannian manifold with five simultaneous emergent properties consistent with both GR and QM. The prior choice is not ad hoc: $\nu = 0.8$ sits in the undefined-mean, power-law regime that is the mathematical signature of quantum foam. The result does not require additional spacetime dimensions, discretisation, or restriction to special geometries.

To our knowledge, no prior work has produced a single probabilistic geometric object simultaneously satisfying GR (Einstein equations, Hawking radiation) and QM (Planck-scale discreteness, wavefunction collapse) from a prior specification alone. Independent replication is invited.

Code and data: https://github.com/RahulModak74/QUANTUM_GRAVITY_WITH_MW

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